

Cycle 16 — Autonomous Agent Report

Date: 2026-05-16

Branch: agent-auto

Repository: qc_exciton_LCC

Cycle goal:

Add a Bloq-driven helper that regenerates the two tabulated dicts the analytic synthesis estimator depends on, so re-tabulation under a QROAMClean upstream change becomes a one-command operation.

Selected deliverable:

- scripts/regenerate_synthesis_tables.py — extracts I_1 and W from the bloq
- tests/test_regenerate_synthesis_tables.py — corner-sample regression + roundtrip

Inbox compliance:

User instruction (2026-05-16): 'send me a pdf report each time, with math equations rendered as latex'. This PDF satisfies that — all math below is rendered via matplotlib mathtext (LaTeX-style) so the report regenerates in any environment with matplotlib (no external LaTeX install required).

Closed-form decomposition (recap)

Toffoli decomposition pinned by tests/test_block_unitary_synthesis_scaling.py:

$$T(M, N, K, b) = K \cdot (2(\log_2 N + 1)b + I_1(M, N))$$

where $M = n_blocks$, $N = n_rows$, $K = n_reflections$, $b = phase_bitsize$.

$I_1(M, N)$ is the per-reflection $b = 0$ intercept tabulated as

SYNTHESIS_PER_REFLECTION_INTERCEPT in model_resource_counts.py.

Qubit decomposition pinned by tests/test_block_unitary_synthesis_qubit_count.py:

$$Q(M, N, b) = [\lceil \log_2 M \rceil + 1 + \log_2 N + b]_{\text{sig}} + W(M, N, b)$$

where W is the transient QROAMClean workspace tabulated as

SYNTHESIS_WORKSPACE_QUBITS.

Sub-linear amortization (cycle 15):

$$T(M)/M \rightarrow 0, \quad T(M) \sim c M^\alpha, \quad \alpha < 1.$$

Empirically (N=256, b=32, K=256): $\alpha_T \approx 0.275$, $\alpha_Q \approx 0.024$.

This cycle does NOT change those identities; it adds a one-command re-tabulation helper that produces both dicts directly from the Bloq.

Helper: scripts/regenerate_synthesis_tables.py

Grid:

$M \in \{1, 2, 4, 8, 16, 32, 64\}$, $N \in \{4, 8, 16, 32, 64, 128, 256\}$

$b \in \{2, 4, 8, 16, 32\}$ (workspace only)

Intercept extraction:

$$I_1(M, N) = T_{\text{bloq}}(M, N, K = 1, b = b_{\text{ref}}) - 2(\log_2 N + 1) b_{\text{ref}}$$

with $b_{\text{ref}} = 4$. Uses QECGatesCost on BlockUnitarySynthesisQROAM.from_shape.

Workspace extraction:

$$W(M, N, b) = \text{QubitCount}(\text{bloq}) - |\text{signature}|$$

Modes:

- check (default) diff regenerated dicts against the shipped tables
- print emit Python source ready to paste back into the module
- no-workspace skip the slower 245-point workspace sweep

Verification run (this cycle):

SYNTHESIS_PER_REFLECTION_INTERCEPT: OK (49 entries match)

SYNTHESIS_WORKSPACE_QUBITS: OK (245 entries match)

Tests & checks

Targeted test runs this cycle:

```
tests/test_regenerate_synthesis_tables.py          4/4 passed (new)
tests/test_block_unitary_synthesis_b_intercept.py 6/6 passed (unaffected)
tests/test_model_resource_counts.py                31/31 passed (unaffected)
scripts/regenerate_synthesis_tables.py --check 2451+49 entries match
```

Achieved goal:

GOALS.md 'Improve constant factors in quantum algorithms' /
'Any potential improvements on the final Toffoli complexity / qubit counts / scaling':
the two tabulated dicts that back the analytic synthesis estimator are now
regenerable from the Bloq with a single command, closing the cycle-12+ recommendation
that re-tabulation under a QROAMClean upstream change should not require a manual edit.

Next recommended task:

Derive $I_1(M, N)$ in closed form from QROAMClean's optimizer. The empirical
asymptotic $I_1(M, N) \sim c\sqrt{MN}$ (consistent with the cycle-15
 $\alpha_T \approx 0.275$ at $N = 256$) and the $n_{\text{blocks}}/n_{\text{rows}}$
monotonicity invariants are the anchors a candidate closed form must satisfy.
The script added this cycle provides a one-command regression check against
the bloq for any candidate closed form.